The leading coefficient for unequal starting exponents

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Abstract

For $p_1 = (h_1+1)/k_1 < p_2 = (h_2+1)/k_2$ the smallest candidate exponent p_1 is a u-pole but no v-pole and no collision, so the Taylor tree has no leading logarithm ($A_{p_1} = 0$) and its leading constant is

$$B_{p_1} = \frac{1}{k_1}\sum_{j\geq 0}\frac{b^{j+\Delta}}{j+\Delta}\,M[p_1;0;c_{0j}], \qquad \Delta = k_2(p_2-p_1) > 0$$

(`leading_coeff_unequal`): the finite part of the leading face is the genuine integral $\int_0^b v^{\Delta-1}a_0(v,s)\,dv$, expanded as an absolutely convergent series (Astra #9 (iii)). Zero `sorry`/`axiom`.

1 The things formalised

```
theorem vExp_ne_uExp_zero_of_lt (h  h  k  k  :  ) (hk  : 0 < k ) (p  p  :  )
    (hp  : ((h  :  ) + 1) / k  = p ) (hp  : ((h  :  ) + 1) / k  = p ) (hlt : p  < p ) (j :  ) :
    vExp h  k  j   uExp h  k  0
theorem leading_coeff_unequal (  b p  p    C  L :  ) (h  h  k  k  D :  ) (h  : 0 <  )
    (hb : 0 < b) (hb  : b <  ) (hk  : 0 < k ) (hk  : 0 < k )
    (hp  : ((h  :  ) + 1) / k  = p ) (hp  : ((h  :  ) + 1) / k  = p ) (hlt : p  < p )
    (c :   ×   →   → ) (hcc :   ij, Continuous (c ij)) (H :   → )
    (hc :   ij s, |c ij s| *   ^ (ij.1 + ij.2)   H s) (hC  : 0   C )
    (henv :   s, 0   s → H s   C  * (1 + s) ^ D * Real.exp (  * s * L)) :
    canonA   h  h  k  k  (fun i j s => c (i, j) s) p  = 0
      canonB   b h  h  k  k  (anaFaceU c b) (anaFaceV c b) (fun i j s => c (i, j) s) p
        = 1 / (k  :  ) *  ' j :  , b ^ ((j :  ) + k  * (p  - p )) / ((j :  ) + k  * (p  - p ))
            * logMoment   p  0 (c (0, j))
```

2 Mathematics

Every v-pole $\delta_j = p_2 + j/k_2 \geq p_2 > p_1 = \alpha_0$, so $C_{p_1} = 0$ and $V_{p_1} = 0$; $B_{p_1} = M[p_1;0;U_{p_1}]$ and by the moment series of unit 126 $M[p_1;0;U_{p_1}] = k_1^{-1}\sum_j q_{\gamma_0}(j)M[p_1;0;c_{0j}]$ with $\gamma_0 = k_2p_1 - h_2 - 1 = k_2(p_1-p_2) = -\Delta < 0$, so no j is resonant and $q_{\gamma_0}(j) = b^{j+\Delta}/(j+\Delta)$.

3 Paper-facing meaning

Complements the equal-start leading coefficient of unit 124: for $p_1 \neq p_2$ the leading term of $Z(\beta,n;\xi,\eta)$ is $n^{-p_1/2}B_{p_1}$ with no logarithm, and B_{p_1} depends on ξ,η along the whole leading face (c_{0j} for all j), not only at the origin. As Astra #9 stressed, this is the correct replacement for "leading exponent p" in the paper's $d = 2$ discussion: the smallest candidate exponent is $\min(p_1,p_2)$, a leading logarithm requires $p_1 = p_2$, and the leading coefficient may vanish.

4 Lean notes

- `rw [← h0]` replaces every occurrence of p_1 by `uExp h  k  0`, including in the target series; rewrite back with `h0` at the end (the final `rw` closes the goal by reflexivity).
- The resonant branch is excluded by `if_neg` from $j \geq 0 > -\Delta$; `sub_neg_eq_add` turns $j - (-\Delta)$ into $j + \Delta$ in both the exponent and the denominator at once.